

MID SEMESTER EXAMINATION MARCH 2025

Name of the Program: B. TECH (Biotech)

Semester: VI

Name of the Course: Fundamentals of Statistics & AI

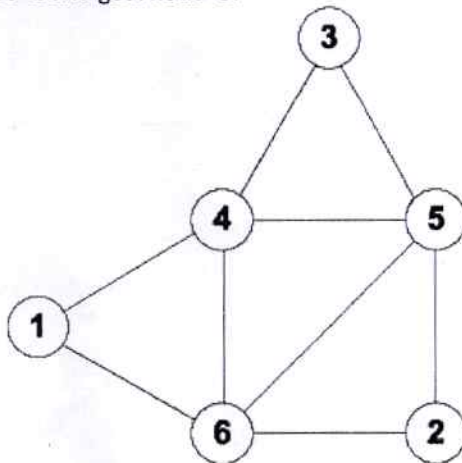
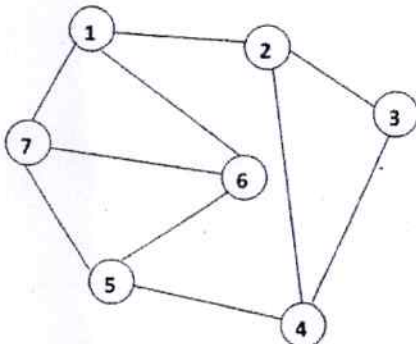
Course Code: TCS 421

Time: 90 minutes

Maximum Marks: 50

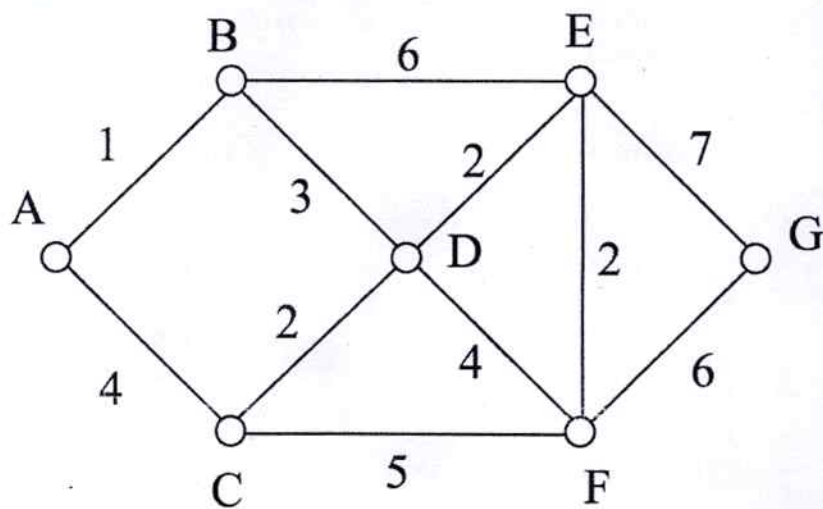
Note:

- (i) Answer **all the questions** by choosing **any one of the sub questions**.
- (ii) Each question carries 10 marks

Q1 (10 marks)		CO1, CO5
a)	Classify the types of Artificial Intelligence and discuss their real time examples.	
OR		
b)	Discuss breadth first search method and apply the same for the given graph. Consider start node=1 and goal node=5.	CO1, CO5
		
Q2 (10 marks)		
a)	Write a short note searching in AI. Identify its types and their suitable use cases.	CO2, CO3
OR		
b)	Explain depth first search method and find the path to reach the goal by applying DFS. Consider start node=1 and goal node=4.	
		CO4, CO5
Q3 (10 marks)		
a)	Make use of AI and Explain few of its possible real time applications.	
OR		

LSH 10

(10 marks)

b)	Solve the following graph to find suitable path to reach goal node using A* technique. Consider start node=A and Goal Node=G. Consider Heuristic values are as follows. <table border="1"><tr><td>A</td><td>B</td><td>C</td><td>D</td><td>E</td><td>F</td><td>G</td></tr><tr><td>8</td><td>6</td><td>6</td><td>4</td><td>2</td><td>2</td><td>0</td></tr></table> 	A	B	C	D	E	F	G	8	6	6	4	2	2	0
A	B	C	D	E	F	G									
8	6	6	4	2	2	0									
Q4 (10 marks)															
a)	Rightly said, "AI is used to develop an intelligent system" Justify this statement in your own words.														
OR															
b)	What is an agent? Explain agents and their role in an Artificially Intelligent System.														
Q5 (10 marks)															
a)	Discuss hill climbing search method with suitable example.														
OR															
b)	Model Data Science Process using a real time example of your own choice domain.														